CDU-1 Atmospheric Distillation Emergency Depressurization Protocol

1. Atmospheric Distillation Column T-101 Process Baseline

2. Critical Emergency Trigger Parameters

Automated Emergency Shutdown ESD-1 triggers automatically when any of the following safety trip limits occur:
1. Fired Heater Coil Outlet Temperature (COT) exceeds 385.0 deg C (detected by transmitters TT-102A/B).
2. Reflux Drum V-101 High-High liquid level reaches 85.0% on level transmitter LT-104 (risk of liquid carryover).
3. Column T-101 Differential Pressure across wash section exceeds 0.45 bar (detected by transmitter PDT-105).
4. Total loss of instrument air header pressure dropping below 4.5 bar gauge.
5. Manual activation of ESD-1 push-button in Main Control Room or Field Marshalling Cabinet.

3. Automated Emergency Shutdown (ESD-1) Execution Sequence

Upon ESD-1 activation, the Triconex safety instrumented system initiates simultaneous protective actions:
a) Immediate fail-close activation of Emergency Shutdown Valve XV-101 on crude charge line 8"-P-101-CS-150.
b) Simultaneous electrical trip of crude charge bottoms pumps P-101A and P-101B.
c) Automated opening of Emergency Blowdown Valve BDV-101, rapidly depressurizing column T-101 vapor inventory into the refinery high-pressure flare header at an engineered controlled rate of 0.8 bar/minute.
d) Primary relief valves PSV-101A and PSV-101B (set at 3.2 bar gauge) provide secondary thermal protection.

CDU-1 Exchanger Bypass & Column Blanketing Safeguards

4. Pre-Heat Exchanger E-101 Emergency Thermal Bypass Protocol

To prevent severe thermal cracking, tube rupture, and localized coking inside pre-heat exchanger E-101 when furnace feed is tripped, operators must execute the emergency bypass protocol:
- Open emergency bypass line 4"-P-103-CS-300 around exchanger E-101 within 45 seconds of feed trip.
- Close motor-operated inlet isolation valve MOV-102.
- Logarithmic Mean Temperature Difference (LMTD) baseline across E-101 under normal operation:
  * Shell Hot Stream (Residue): T_hot_in = 160.0 deg C, T_hot_out = 120.0 deg C
  * Tube Cold Stream (Crude): T_cold_in = 30.0 deg C, T_cold_out = 85.0 deg C
  * Delta T1 = 160 - 85 = 75.0 deg C | Delta T2 = 120 - 30 = 90.0 deg C
  * Verified LMTD = (90.0 - 75.0) / ln(90.0 / 75.0) = 82.26 deg C (Vendor claim: 82.0 deg C, Discrepancy: 0.31%).

5. Post-Trip Column Inerting and Vacuum Prevention

As column T-101 cools post-trip, condensing hydrocarbon vapor creates severe negative pressure (vacuum risk). To prevent catastrophic tray deformation and structural shell implosion:
1. Open utility nitrogen blanketing valve N2-PCV-108 immediately when overhead pressure drops below 0.6 bar.
2. Maintain positive nitrogen pressure inside column T-101 strictly above 0.35 bar gauge at all times.
3. Never open column manways or vents to atmospheric air until internal oxygen concentration is verified < 0.5%.